



CODECELL - GMPN, VESIT PRESENTS
IN ASSOCIATION WITH UIC & IQAG



SYRUS 2026

No checkpoints. No retries. Just the heist.

PS-03: Autonomous Developer Onboarding Agent

Background

Onboarding a new developer is one of the most resource-intensive processes in a software organization. The hard part? It requires coordinating multiple tools, systems, and stakeholders just to get a new hire productive. Developers must gain access to repositories, set up tools, read documentation, understand architecture, learn internal processes, and complete compliance tasks. While companies use platforms like GitHub, Slack, Jira, and documentation systems like Notion or Confluence, the onboarding process still relies heavily on manual coordination. This leads to delayed productivity, repeated explanations, inconsistent onboarding experiences, incomplete checklists, and poor visibility into onboarding progress. We need a smarter onboarding system — an intelligent agent that can autonomously guide new hires, personalize onboarding paths, track progress, and notify HR upon successful completion.

Problem Statement

Organizations lack an autonomous onboarding system that can guide new software developers through company processes, personalize their onboarding experience based on role and experience, track completion, and generate structured confirmation to HR — all using a structured company knowledge base.

Objective

Build an Advanced Agentic Onboarding Platform that:

1. Guides new employees step-by-step via a chat-based interface (Mandatory)
2. Personalizes onboarding based on:
 - Role (Backend / Frontend / DevOps)
 - Experience Level (Intern / Junior / Senior)
 - Tech Stack (Python / Node / Java, etc.)
3. Uses a provided structured knowledge base with retrieval (RAG-style)
4. Completes an onboarding checklist
5. Sends a structured completion email to HR (Mandatory)



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Target Users

- New Interns
- Software Engineers (Junior to Senior)
- HR Teams
- Engineering Managers
- DevOps Teams

Expected Solution Features

Your solution should include:

- Personalized onboarding flow: Ask structured onboarding questions, identify the employee persona (role, tech stack, experience level), and dynamically generate a customized onboarding path.
- Knowledge base integration: Retrieve relevant company documentation such as policies, setup guides, and engineering standards; answer contextual queries accurately while referencing source documents and avoiding hallucinated information.
- Checklist tracking: Maintain a structured onboarding checklist, track completed tasks, and clearly indicate onboarding status (In Progress / Completed).
- HR notification system: Upon completion, send a structured summary email including employee name, role and team, completed items, pending tasks (if any), timestamp of completion, and an optional confidence score.

Constraints

- Must implement chat-based onboarding interface
- Must generate structured HR email
- Should avoid hallucination beyond knowledge base
- Focus on agent intelligence over number of integrations
- Integration APIs may be mocked



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Example Workflows & Interactions

- Example Interaction:
User: “Hi, I’m Riya. I’ve joined as a Backend Intern working on Node.js.”
Agent should: Identify persona, provide setup steps, share backend docs, guide environment setup, provide a small starter task, mark the checklist complete, and send the HR completion email.
- Personalized Onboarding Flow Examples:
 - Backend Intern (Node.js): Set up Node environment → Clone repository → Run local server → Review backend architecture doc → Read API standards → Complete starter bug fix.
 - Frontend Senior (React): Review frontend architecture → Access design system → Understand deployment pipeline → Review PR guidelines.

Bonus (Optional)

- Provision or request access for GitHub, Slack, and Jira.
- Send an automated Slack welcome message.
- Assign Jira or GitHub starter tickets.
- Automatically verify local environment setup.
- Track checklist completion dynamically within the chat.
- Update the internal knowledge base with new FAQs generated during onboarding.
- Ingest PDFs instead of Markdowns in the vectorDB to align more with practical use.
- Use Model Context Protocol (MCP) style architecture for modular design.

Evaluation Criteria

1. Agent Intelligence (30%) – Personalization quality , logical onboarding sequencing, context awareness
2. Knowledge Retrieval Quality (20%) – Accurate document referencing, minimal hallucination
3. System Architecture (20%) – Modular design, agent workflow clarity, scalability
4. Completion Reporting (10%) – Professional HR email, structured reporting
5. Innovation – Novel interaction design, adaptive learning behavior
6. Bonus Integrations (10%) – GitHub / Slack / Jira automation , MCP usage



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Deliverables

- Working prototype
- 5-7 minute demo video
- Architecture diagram
- Presentation (max 10 slides)
- Sample onboarding walkthrough (Intern → Completed → HR email)

Provided Dataset/ Resources: [Syrus2026-Resources/PS03](#)